SYNOPSIS REPORT

ON

SMART FINDER AND RECOVERY MANAGEMENT SYSTEM

Submitted by:

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2025-2026

Certificate of Project Confirmation

This is to certify that the Project title: " SMART FINDER AND RECOVERY MANAGEMENT SYSTEM" **assigned to group TYCO as a part of fulfillment of Diploma in Computer Engineering on 15 day of October 2025.**

Members of project group

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The group will be complete the project within the academic schedule under the guidance of

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Project Synopsis

Project Title: SMART FINDER AND RECOVERY MANAGEMENT SYSTEM

1. Introduction

In today's rapidly advancing digital world, the importance of technology in solving real-life problems cannot be overstated. One common yet overlooked issue is the **loss of valuable items or missing people**. Every day, across schools, offices, transport systems, and public gatherings, people lose belongings such as mobile phones, wallets, or documents, and in some unfortunate situations, people themselves go missing.

Traditional approaches to recovery—like manual notices, police reports, or announcements—are slow, unorganized, and geographically limited. To address this problem, the **Smart Finder and Recovery Management System** offers a **centralized digital platform** that connects people who lose something with those who find it, including cases of missing individuals.

This system uses **Django (Python)** for the backend and **MongoDB** as a NoSQL database, ensuring scalability and flexibility. The project enables users to report **lost items or missing persons**, upload descriptions and images, and search through verified reports using location, category, and keywords.

The aim is to create an **intelligent, user-friendly, and secure** web portal that simplifies the process of recovering lost items and people through automation, data management, and online connectivity. This project not only demonstrates technical skill but also social awareness and innovation.

2 PROBLEM STATEMENT

Despite the availability of modern communication systems, recovering lost items or tracing missing individuals still depends on manual, time-consuming efforts. The absence of a **centralized, trustworthy digital system** often leads to wasted time, duplicated efforts, and unclaimed items.

Major problems include:

- No single platform to record lost and found cases.
- Slow response and lack of awareness.
- Difficulty verifying information authenticity.
- Poor communication between finders, owners, and authorities.

In case of missing people, traditional approaches such as posters or announcements reach a very limited audience. Similarly, valuable objects often remain unclaimed or end up in storage due to lack of visibility.

Hence, there is a need for a **web-based digital platform** that automates and centralizes this entire process. The **Smart Finder and Recovery Management System** addresses this by allowing users to report, track, and recover both items and people quickly and securely.

3. OBJECTIVES

- To develop an intelligent online portal for lost and found management of both **items and people**.
- To maintain a secure, centralized database using Django and MongoDB.
- To enable users to register and post detailed lost or found reports with images.
- To provide **smart search and automatic matching** between reports.
- To allow administrators to verify and manage records.
- To improve transparency, speed, and reliability in the recovery process.
- To encourage public participation and awareness in digital recovery systems.

4. SCOPE / FEATURES

User Features

- User registration and authentication.
- Option to post reports for lost items or missing people.
- Upload photos, contact details, date, and location of loss/found.
- Smart search and auto-match suggestions.
- Receive notifications when a potential match is found.

Admin Features

- Verify and approve user-submitted reports.
- Manage database integrity and flag false entries.
- Generate statistical reports of active cases.

System Features

- Multi-role access (User/Admin).
- Secure login and data encryption.
- Responsive design accessible via mobile or desktop.
- Notification system for matched records.
- Real-time updates and tracking of report status.

5. SYSTEM DESIGN & METHODOLOGY

Architecture

The project uses a **three-tier architecture**:

1. **Presentation Layer (Frontend):** Built with HTML, CSS, Bootstrap, and JavaScript.
2. **Application Layer (Backend):** Django manages logic, routing, and API handling.
3. **Database Layer:** MongoDB stores user details, reports, and matches.

Modules

1. **User Module:** Handles registration, login, and user profile management.
2. **Lost/Found Module:** Records items or people with images and location.
3. **Matching Module:** Compares lost and found entries to suggest possible matches.
4. **Admin Module:** Manages approvals and ensures authenticity.
5. **Notification Module:** Alerts users when a match occurs.

Methodology

The **Agile development approach** was used:

- **Planning:** Identify the system's goals and requirements.
- **Design:** Create database schema, data flow diagrams, and UI mockups.
- **Development:** Code in Django and connect MongoDB using Djongo.
- **Testing:** Unit and integration testing.
- **Deployment:** Host using GitHub or Render platform.
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6. EXPECTED OUTCOMES

- Centralized digital platform for reporting lost and found items/people.
- Reduction in time to locate lost objects or missing individuals.
- Transparent and verified data management.
- Scalable architecture for future improvements.
- Increased social awareness and participation.
- Integration-ready design for AI, GPS, or mobile application in future.

7. TOOLS AND TECHNOLOGIES

The **Smart Finder and Recovery Management System for Lost Items and Missing People** is developed using a combination of modern web technologies that ensure scalability, performance, and user-friendliness. Each component of the system—from the frontend interface to the backend logic and database—has been carefully chosen to meet the project's functional and non-functional requirements. The technologies used are open-source, platform-independent, and easy to maintain, making the system reliable and efficient for long-term use.

Frontend Technologies

The frontend or client-side of the application is developed using **HTML (Hypertext Markup Language)**, **CSS (Cascading Style Sheets)**, and **JavaScript**.

- **HTML5** provides the structure of the web pages and defines how content such as text, images, and forms is presented to users. It also supports responsive and semantic elements which help in designing a clean and accessible user interface.
- **CSS3** is used for styling the website, enhancing its visual appeal through layouts, color schemes, animations, and responsive designs. The use of CSS ensures that the website maintains a consistent appearance across various devices such as desktops, tablets, and smartphones.
- **JavaScript** adds interactivity to the application. It is used to validate user input, dynamically update content without reloading pages, and enhance the user experience

with features such as live search and instant feedback.
Additionally, **Bootstrap**, a powerful front-end framework, is utilized to make the

design mobile-friendly and visually appealing. Bootstrap's predefined templates and grid system help create responsive layouts with minimal effort, ensuring the system can adapt to different screen sizes seamlessly.

Backend Technologies

The backend is the heart of the system, responsible for processing user requests, managing data, and ensuring the logical flow of operations. The project uses **Django**, a high-level Python web framework, to manage all server-side functionalities. Django follows the **Model-View-Template (MVT)** architectural pattern, which separates business logic from user interface design, resulting in clean, maintainable, and scalable code.

Some of the major advantages of Django include:

- **Built-in authentication and security** features like password hashing, session management, and cross-site request forgery (CSRF) protection.
- **Automatic admin interface**, which allows administrators to monitor and manage the system effortlessly.
- **ORM (Object Relational Mapper)**, which simplifies database interactions and eliminates the need to write complex SQL queries.
- **Rapid development capabilities**, allowing faster implementation of complex functionalities such as report matching and user management.

Django's reliability and support for multiple databases make it an ideal choice for this project. It also integrates well with external APIs and modules, allowing the future expansion of the system into areas such as AI-based matching or cloud-based notifications.

Database Technology

The project uses **MongoDB**, a powerful **NoSQL database**, to store all the system's data. Unlike traditional relational databases, MongoDB stores data in a **flexible JSON-like format (BSON)**, which makes it ideal for handling complex and unstructured data such as images, descriptions, and location details of lost items or people.

MongoDB was selected because of its ability to scale horizontally, handle large amounts of data efficiently, and provide high-speed data retrieval. The connection between Django and MongoDB is made using **Djongo**, a connector that allows Django's ORM to communicate with MongoDB collections as if they were relational tables.

The advantages of MongoDB in this project include:

- High performance for read/write operations.
- Dynamic schema, allowing different types of reports to be stored without structural limitations.
- Easy integration with Python and cloud platforms like MongoDB Atlas.
- Secure data storage with authentication and user-level access control.

This choice of database makes the system highly adaptable for future extensions such as AI-based search algorithms or cloud-based backups.

Development and Deployment Tools

The project is developed using **Visual Studio Code (VS Code)**, an open-source integrated development environment (IDE) known for its simplicity and extensive support for Python and web development. It provides features like syntax highlighting, debugging tools, and Git integration that help in managing and maintaining the code effectively.

Version control is maintained using **Git**, and the project repository is hosted on **GitHub**. This allows for collaborative development, version tracking, and public sharing of the project. Using Git ensures that changes in the source code are tracked efficiently, and older versions can be restored if necessary.

For local testing, Django's built-in **development server** is used, which enables the project to be executed on localhost before deployment. For demonstration and online accessibility, the system can be deployed using platforms like **Render** or **PythonAnywhere**, which support Python and Django applications without additional cost.

Operating System and Environment

The project is compatible with multiple operating systems, including **Windows**, **Linux**, and **macOS**, since both Django and MongoDB are cross-platform. The system requires Python 3.x installed on the host machine and a browser such as Google Chrome or Mozilla Firefox to run the web interface.

Hardware requirements are minimal: an Intel i3 or higher processor, 4 GB of RAM, and around 500 MB of free disk space for installation and data storage. This makes the system affordable and easily deployable even on personal laptops or institutional computers.

Summary

In summary, the project leverages a combination of **modern, reliable, and open-source technologies** to achieve its objectives.

- The **frontend** ensures usability and responsiveness.
- The **backend** handles logical operations, authentication, and secure data flow.
- The **database** ensures flexible and efficient storage of all reports.
- The **development and deployment tools** provide a smooth workflow from coding to hosting.

Together, these technologies form a powerful, efficient, and scalable foundation for the **Smart Finder and Recovery Management System**, making it capable of addressing real-world problems through a modern technological approach.

8. LIMITATIONS

Although the **Smart Finder and Recovery Management System** provides an efficient and user-friendly platform for managing lost and found cases, it has some limitations in its current version.

The system primarily depends on an **active internet connection**, which limits its accessibility in remote areas with poor connectivity. Users must be online to report or search for lost and found cases, as no offline functionality is yet provided.

Another limitation is that the **accuracy of data** relies on the correctness of the information entered by users. If users provide incomplete or incorrect details, the system may fail to identify accurate matches. At present, the matching process is based mainly on **textual and location-based comparisons**, and advanced AI or image recognition features are not yet implemented.

The **security features** in the current version are basic, using Django's built-in authentication and encryption. However, it does not yet support OTP verification or two-factor authentication for added safety.

The system also lacks **integration with external or government databases** such as police or transport authorities, which could further improve the effectiveness of tracking missing persons and unclaimed items.

Lastly, GPS-based **real-time location tracking** and a **mobile application** are not yet included. These can be added in future enhancements to make the system more accessible and intelligent.

Despite these limitations, the project effectively demonstrates the concept of a centralized digital solution for managing lost items and missing people and provides a strong base for future improvements.

9. CONCLUSION

The **Smart Finder and Recovery Management System for Lost Items and Missing People** is a practical and innovative web-based solution designed to bridge the gap between those who lose something and those who find it. By integrating modern web technologies such as **Django** and **MongoDB**, the system provides a reliable, secure, and user-friendly platform for reporting, searching, and recovering lost items or missing individuals.

This project successfully demonstrates how technology can simplify real-world problems by automating manual and time-consuming tasks. It ensures **data transparency, quick communication, and efficient record management**, reducing dependency on traditional methods like notices or manual reporting.

Although the current version has certain limitations, such as the absence of AI-based image matching and GPS tracking, it establishes a **strong foundation** for future development. Features like mobile application support, government database integration, and advanced verification systems can enhance its scope further.

In conclusion, the system not only showcases technical proficiency but also emphasizes **social responsibility** by helping people recover their valuables and reconnect with missing individuals through the power of technology.

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